LIBBY — TARGET VALIDATION PATHS

Target validation paths — osteosarcoma-mets-dll3-h7r2

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

The case has two RNA-only targetable features, each with its own protein-level confirmation step. **DLL3 IHC (clone SP347)** gates the DLL3-directed pathway: tarlatamab via NCT06788938 and the SHR-4849 / IDE849 ADC trial NCT07174583. **PRAME IHC plus HLA-A*02:01 typing** together gate the PRAME-directed pathway: IMA203 via NCT03686124 (Immatics ACTengine) and IMC-P115C via NCT07156136. The two pathways are independent. The patient may have neither, either, or both. All three workup tests can run in parallel on archival tissue and a single blood draw. If both workups return negative, this report has no within-scope recommendations, and the next conversation about standard 2L+ care is the treating team's, not Libby's.

DLL3

Before any DLL3-directed therapy decision: DLL3 IHC SP347 on archival FFPE, ≥1% (preferably ≥25%) per NCT06788938's enrollment threshold. Turnaround is one to three weeks; cost is trivial relative to a treatment cycle. First, confirm SP347 assay availability at the treating institution. Not every reference lab carries the Roche Tissue Diagnostics clone used in the tarlatamab development program.

Two refinements sit one tier down. Spatial heterogeneity is a known confounder in solid-tumor DLL3 (Zhang 2023): if multiple tumor blocks are available, IHC on a metastatic site as well as the primary refines confidence in whether the gating result generalizes. Neuroendocrine context (ASCL1 / NEUROD1 / chromogranin / synaptophysin / INSM1 panel) is research-grade for an osteosarcoma. DLL3 is normally a Notch-pathway target on neuroendocrine lineage, so an unexpectedly positive DLL3 IHC in a non-NEC tumor is worth contextualizing before the trial enrollment paperwork.

Germline TP53 sequencing (Li-Fraumeni panel) is a separate kind of finding. It does not affect tarlatamab eligibility, but late-teens / twenties osteosarcoma carries a meaningful prior probability for germline TP53. A positive result changes radiation-sensitivity planning, prompts screening for synchronous tumors, and triggers cascade testing for first-degree relatives. Discuss with a genetic counselor before ordering.

PRAME

Two essential tests gate the entire PRAME-directed pathway: **PRAME IHC** confirms tumor protein expression, and **HLA-A*02:01 typing** confirms the patient can present the PRAME peptide. Every PRAME-directed ImmTAC and TCR-T in clinical development (IMA203, brenetafusp, IMC-P115C) is HLA-A*02:01-restricted; typing-negative patients are foreclosed even when PRAME IHC is strongly positive. PRAME IHC is positive in roughly 56% of osteosarcoma (Iura 2017, n=82); HLA-A*02:01 prevalence is ~40–50% in Caucasian populations. Order both alongside the DLL3 IHC: same block, same blood draw, same turnaround window.

Spatial heterogeneity refines confidence at the next tier. PRAME IHC on a metastatic site as well as the primary helps rule out the case where one biopsy hits a focal expression hotspot. Tumor NGS for B2M loss-of-function and HLA-A*02:01 loss-of-heterozygosity is a resistance check: TCR-mediated killing depends on intact antigen presentation, and tumor-specific HLA-LOH is a documented escape mechanism that the IHC and HLA-typing workup will miss by design. Quantifying baseline CD8 T-cell infiltrate density via IHC (or multiplex CD3 / CD8 / FoxP3) is the microenvironment companion: an immune-cold tumor predicts diminished response even when PRAME and HLA are both positive.

Where to order these assays

The preferred provider for each assay is marked **(preferred)**, selected on company size, reputation, US-based location, and turnaround time. Other providers in the row are listed in case the preferred lab is unreachable for this patient.

Assay	Provider	Decision gated	Contact
DLL3 IHC (clone SP347)	**Foundation Medicine (preferred) (FoundationOne CDx + IHC reflex)**	**Tarlatamab via NCT06788938**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
DLL3 IHC (clone SP347)	Mayo Clinic Laboratories	Tarlatamab via NCT06788938	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
DLL3 IHC (clone SP347)	NeoGenomics Laboratories	Tarlatamab via NCT06788938	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
DLL3 IHC (clone SP347)	Caris Life Sciences	Tarlatamab via NCT06788938	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
DLL3 IHC (clone SP347)	LabCorp / Esoterix Oncology	Tarlatamab via NCT06788938	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
DLL3 multi-region IHC (heterogeneity)	**NeoGenomics Laboratories (preferred)**	**Refines DLL3 IHC gating result; does not gate enrollment**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
DLL3 multi-region IHC	Foundation Medicine	Refines DLL3 IHC gating result; does not gate enrollment	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
DLL3 multi-region IHC	Mayo Clinic Laboratories	Refines DLL3 IHC gating result; does not gate enrollment	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
DLL3 multi-region IHC	Memorial Sloan Kettering Diagnostic Molecular Pathology	Refines DLL3 IHC gating result; does not gate enrollment	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Neuroendocrine IHC panel (ASCL1 / NEUROD1 / chromogranin / synaptophysin / INSM1)	**Mayo Clinic Laboratories (preferred)**	**Confirms DLL3 target call (neuroendocrine biology); does not gate enrollment**	**test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710**
Neuroendocrine IHC panel	ARUP Laboratories	Confirms DLL3 target call (neuroendocrine biology); does not gate enrollment	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787
Neuroendocrine IHC panel	LabCorp / Esoterix Oncology	Confirms DLL3 target call (neuroendocrine biology); does not gate enrollment	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
Neuroendocrine IHC panel	Quest Diagnostics	Confirms DLL3 target call (neuroendocrine biology); does not gate enrollment	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378

Neuroendocrine IHC panel	Memorial Sloan Kettering Diagnostic Molecular Pathology	Confirms DLL3 target call (neuroendocrine biology); does not gate enrollment	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Germline TP53 (Li-Fraumeni panel)	**Invitae (preferred) (Common Hereditary Cancers Panel)**	**Reframes radiation, synchronous-tumor screening, and family cascade testing; does not affect tarlatamab eligibility**	**test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037**
Germline TP53 panel	GeneDx	Reframes radiation, synchronous-tumor screening, and family cascade testing; does not affect tarlatamab eligibility	test info · 207 Perry Parkway, Gaithersburg, MD 20877 · 1-888-729-1206
Germline TP53 panel	Ambry Genetics (CancerNext)	Reframes radiation, synchronous-tumor screening, and family cascade testing; does not affect tarlatamab eligibility	test info · 1 Enterprise, Aliso Viejo, CA 92656 · 1-866-262-7943
Germline TP53 panel	Myriad Genetics (MyRisk Hereditary Cancer)	Reframes radiation, synchronous-tumor screening, and family cascade testing; does not affect tarlatamab eligibility	test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423
Germline TP53 panel	Color Health	Reframes radiation, synchronous-tumor screening, and family cascade testing; does not affect tarlatamab eligibility	test info · 831 Mitten Road, Burlingame, CA 94010 · 1-844-352-6567
PRAME IHC (clone EPR20330 or equivalent)	**NeoGenomics Laboratories (preferred)**	**IMA203 (NCT03686124) and the PRAME ImmTAC class (brenetafusp, IMC-P115C); every program gates on PRAME IHC**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
PRAME IHC	Caris Life Sciences	IMA203 / brenetafusp / IMC-P115C; every program gates on PRAME IHC	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
PRAME IHC	Foundation Medicine	IMA203 / brenetafusp / IMC-P115C; every program gates on PRAME IHC	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
PRAME IHC	LabCorp / Esoterix Oncology	IMA203 / brenetafusp / IMC-P115C; every program gates on PRAME IHC	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
PRAME IHC	Mayo Clinic Laboratories	IMA203 / brenetafusp / IMC-P115C; every program gates on PRAME IHC	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
HLA Class I high-resolution typing (HLA-A*02:01)	**HistoGenetics (preferred)**	**IMA203 and the PRAME ImmTAC class; every program is HLA-A*02:01-restricted**	**test info · 1 Patrick Henry Drive, Stewartsville, NJ 08886 · 1-845-356-3801**
HLA Class I typing	Versiti / Wisconsin Diagnostic Laboratories	IMA203 and the PRAME ImmTAC class; every program is HLA-A*02:01-restricted	test info · 638 N 18th Street, Milwaukee, WI 53233 · 1-800-245-3117

HLA Class I typing	ARUP Laboratories	IMA203 and the PRAME ImmTAC class; every program is HLA-A*02:01-restricted	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787
HLA Class I typing	Mayo Clinic Laboratories	IMA203 and the PRAME ImmTAC class; every program is HLA-A*02:01-restricted	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
HLA Class I typing	Stanford Histocompatibility Laboratory	IMA203 and the PRAME ImmTAC class; every program is HLA-A*02:01-restricted	test info · 3373 Hillview Avenue, Palo Alto, CA 94304 · 1-650-723-7960
PRAME multi-region IHC (heterogeneity)	**NeoGenomics Laboratories (preferred)**	**Refines PRAME IHC gating result; does not gate enrollment**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
PRAME multi-region IHC	Caris Life Sciences	Refines PRAME IHC gating result; does not gate enrollment	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
PRAME multi-region IHC	Foundation Medicine	Refines PRAME IHC gating result; does not gate enrollment	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
PRAME multi-region IHC	Mayo Clinic Laboratories	Refines PRAME IHC gating result; does not gate enrollment	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
PRAME multi-region IHC	Memorial Sloan Kettering Diagnostic Molecular Pathology	Refines PRAME IHC gating result; does not gate enrollment	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Tumor NGS for HLA-LOH / B2M / antigen-presentation	**Memorial Sloan Kettering (preferred) (MSK-IMPACT)**	**Frames durability and post-progression sequencing for PRAME-directed therapy**	**test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000**
Tumor NGS (HLA-LOH / B2M)	Foundation Medicine (FoundationOne CDx)	Frames durability and post-progression sequencing for PRAME-directed therapy	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor NGS (HLA-LOH / B2M)	Caris Life Sciences (Molecular Intelligence)	Frames durability and post-progression sequencing for PRAME-directed therapy	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
Tumor NGS (HLA-LOH / B2M)	Tempus (xT)	Frames durability and post-progression sequencing for PRAME-directed therapy	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Tumor NGS (HLA-LOH / B2M)	NeoGenomics Laboratories (NeoTYPE Comprehensive)	Frames durability and post-progression sequencing for PRAME-directed therapy	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
CD8 / TIL density (multiplex IHC)	**NeoGenomics Laboratories (preferred) (MultiOmyx)**	**Frames durability of PRAME-directed therapy; informs the next-line plan if IMA203 or IMC-P115C fails**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
CD8 / TIL density	Caris Life Sciences	Frames durability of PRAME-directed therapy	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669

CD8 / TIL density	Foundation Medicine	Frames durability of PRAME-directed therapy	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
CD8 / TIL density	Mayo Clinic Laboratories	Frames durability of PRAME-directed therapy	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
CD8 / TIL density	Memorial Sloan Kettering Diagnostic Molecular Pathology	Frames durability of PRAME-directed therapy	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements
DLL3 IHC (clone SP347)	RNA expression establishes that the DLL3 gene is being transcribed; it does not establish that DLL3 protein sits on the cell surface where a BiTE can engage. NCT06788938 enforces IHC ≥ 25% (stage 1) or ≥ 1% (stage 2) for enrollment, and every approved DLL3-directed therapy gates on protein-level confirmation. Skipping the test forecloses every DLL3-directed candidate downstream by definition.	Foundation Medicine (FoundationOne CDx (DLL3 IHC reflex)) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	archival FFPE; fresh biopsy not required if stored tissue is adequate
DLL3 IHC on multiple tumor regions / metastatic biopsy	Osteosarcoma metastases can diverge from primary tumors in surface-marker expression, and the cross-tumor DLL3 literature in solid tumors (Zhang 2023) flags spatial heterogeneity as a frequent confounder. Testing one site can over- or under-estimate enrollment-grade DLL3 status. If only one block is available the workup proceeds with that block, but the result should be interpreted with this caveat in mind.	NeoGenomics Laboratories · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	additional archival blocks from a different site if available

ASCL1 / NEUROD1 IHC + chromogranin / synaptophysin / INSM1 panel	DLL3 is a Notch-pathway target normally expressed in neuroendocrine lineage. In a non-NEC tumor like osteosarcoma, surfacing or excluding any neuroendocrine differentiation pattern (Notch-low / ASCL1-high state) provides mechanistic context for whether DLL3 expression is biologically plausible or a stochastic finding. Does not gate enrollment; informs how the tarlatamab trial outcome should be interpreted.	Mayo Clinic Laboratories · test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710	archival FFPE; same block as the DLL3 IHC
Germline TP53 sequencing (Li-Fraumeni panel)	Osteosarcoma in the late-teens to twenties carries a meaningful prior probability of germline TP53 (Li-Fraumeni). A positive germline TP53 result changes radiation-sensitivity considerations, screening for synchronous tumors, and family screening for first-degree relatives. Does not affect tarlatamab eligibility, but is the kind of finding that would change the surrounding care plan.	Invitae (Invitae Common Hereditary Cancers Panel) · test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037	5–10 mL EDTA whole blood

PRAME IHC (clone EPR20330 or equivalent)	RNA expression establishes that the PRAME gene is being transcribed; it does not establish that the PRAME peptide is presented at sufficient density to engage a TCR or ImmTAC. PRAME-targeting therapies in development (IMA203 ACTengine, brenetafusp, IMC-P115C) require IHC-confirmed PRAME protein expression on tumor cells. PRAME IHC is positive in ~56% of osteosarcoma (Iura 2017, n=82). Skipping the IHC forecloses every PRAME-directed candidate downstream by definition.	NeoGenomics Laboratories · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; same block as the DLL3 IHC
HLA Class I high-resolution typing (HLA-A*02:01 specifically)	Every PRAME-directed ImmTAC and TCR-T in clinical development is HLA-A02:01-restricted; the PRAME peptide is presented on HLA-A02:01, and patients without this allele cannot present the target. HLA-A02:01 prevalence is ~40–50% in Caucasian populations; without typing, the patient's eligibility for the entire HLA-restricted PRAME class is unknown. Every PRAME-directed program (IMA203 via NCT03686124, brenetafusp, IMC-P115C via NCT07156136) is foreclosed if HLA-A02:01 typing returns negative.	HistoGenetics · test info · 1 Patrick Henry Drive, Stewartsville, NJ 08886 · 1-845-356-3801	5–10 mL EDTA whole blood

PRAME IHC on multiple tumor regions / metastatic biopsy	PRAME expression is heterogeneous across tumor regions, particularly in solid tumors outside melanoma. Iura 2017 reported variable PRAME staining intensity across osteosarcoma cases; primary-vs-metastatic discordance is a documented confounder for PRAME-directed therapy selection. Multi-region IHC refines confidence that the binary positive call is representative of the dominant disease.	NeoGenomics Laboratories · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	additional archival blocks from a different site if available
Tumor NGS for HLA B2M loss-of-function, and pathway	TCR-based and ImmTAC therapies depend on intact HLA-Class-I antigen presentation on tumor cells. Tumor-specific HLA-A*02:01 loss-of-heterozygosity, B2M loss-of-function mutations, or upstream antigen-presentation defects render the cell invisible to TCR-mediated killing even with PRAME expression preserved. Detecting these alterations does not foreclose enrollment but reframes the durability expectation and informs sequencing planning.	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT) · test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000	archival FFPE; matched germline blood is required for tumor-specific HLA-LOH calling
Tumor CD8+ T-cell infiltrate density (CD8 IHC, optionally with multiplex CD3 / CD8 / FoxP3)	TCR-based therapies require T-cell trafficking into the tumor; an immune-cold microenvironment with low CD8 density predicts diminished response even when PRAME and HLA are both positive. Quantifying baseline TIL density frames durability expectations and is informative for the next-line conversation if PRAME-directed therapy fails. Does not gate enrollment.	NeoGenomics Laboratories (MultiOmyx multiplex IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; same block as the PRAME IHC

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.